

**MODULAR CONSTRUCTION: DEVELOPING A BIM
BASED DESIGN EVALUATOR FOR BUILDING
COMPONENTS**

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thesis titled
**MODULAR CONSTRUCTION: DEVELOPING A BIM
BASED DESIGN EVALUATOR FOR BUILDING
COMPONENTS**

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***Dedicated
To
my Family***

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LIST OF ABBRIVIATIONS

API	Application Programming Interface
BIM	Building Information Modelling
BIM-DE4PB	BIM-Design Evaluation for Prefabricated
CAD	Computer Aided Designs
DFA	Design for Assembly
DFM	Design for Manufacture
DFMA	Design for Manufacturing and assembly
FIDIC	International Federation for Consulting Engineers
GCC	General Conditions of Contract
ICT	Information Communication Technology
IDSS	Integrated decision support system
LCP	Lean Construction Principles
LPS	Last Planner system
PSOM	Particle Swarm Optimization Model
RII	Relative Importance Index
ROI	Return of Investment
SCE	Small Construction Enterprises

ABSTRACT

The paucity of appropriate consideration of factors in designing modular construction elements will result a multitude of disparities in the form of aggravated cost, men power, safety problems, construction delays and inefficient material selection (Manu et al., 2019). Fragmented nature of construction industry has made it more complex to understand. Increasing complexity of prefabricated construction is one of the leading sources of project failure (Davila Delgado et al., 2017). Customary methods of construction generate more material waste during execution, inefficient design, more time requirement, congested workspace, untrained labor work, less quality control and weather dependency (Jin et al., 2018). It is important to make guided decisions at early stage of design phase to increase production efficiency during construction (Boothroyd, G., P. Dewhurst, 2001). The fundamental purpose of lean construction principle is waste elimination from preliminary stage of construction and make the whole process simple (Aziz & Hafez, 2013). Design for manufacture (DFM) and design for assembly (DFA) are two techniques used to simplify the whole product structure, to reducing manufacturing and assembly cost and minimize the waste material at early design conception stage (Boothroyd, G., P. Dewhurst, n.d.). To address these problems, this study aims to integrate Design for Manufacture and Assembly (DFMA) and lean construction principles in Building Information Modeling (BIM) (Manu et al., 2019). By integrating the above techniques, this study will propose a framework for design evaluation through BIM. The assessment and evaluation method will rely on DFMA, lean construction principles and expert's opinion. This framework may help designers to optimize their design from different available alternative options at preliminary design stage of different building elements. The integrated framework may also help to develop a BIM based digital platform for assessment and evaluation of prefabricated building elements for different sectors of construction industry during the early design stage. The BIM based digital platform help clients by giving them an overview of available better options for their needs by implementing it at preliminary stage of construction. Resultantly efficiency of whole construction processes may increase (Manu et al., 2019).

INTRODUCTION

1.1 GENERAL

The quality of the final product of design in civil construction industry has great importance but it has been observed that little importance is given to the design development stage and conception stage. When it comes to construction of prefabricated building elements it becomes critical. During the early stage of design of prefabricated construction, it is very important to make a guided decision to enhance the efficiency of the final product. DFMA has successfully applied in manufacturing industry to improve the design attributes such as design flexibility, cost of the design, environmental suitability, easy of manufacturing and material sourcing. We can evaluate different design option at the early stage of design using DFMA. Prefabricated construction involved both manufacturing and transportation of building elements at construction site. Although manufacturing industry is very efficient than construction industry, but attempt can be made for evaluation the design of prefabricated building elements to incorporate the efficiency in construction industry. It can be done through standardization, preassembly engineering modifications, transportation, and lean principles incorporation. According to Lean construction principles have harmonious relationship with design for manufacture and assembly through improving design optimization, reducing construction waste at initial stage. Traditionally design, is developed by several stakeholders such as architects, engineers of different specialties but they do not have proper coordinated platform to evaluate their design at conception stage of design. The BIM based integration of DMFA and lean principles to evaluate the design of prefabricated building elements has great potential for the promotion of efficient prefabricated construction.

1.2 PROBLEM STATEMENT

The construction industry has been successfully adopting manufacturing industry techniques to increase its efficiency. It has greatly influenced the whole construction process (Aziz & Hafez, 2013). The inefficiencies in construction operations are significantly reduced through improvement of construction process and waste minimization for different building elements (Aziz & Hafez, 2013). DFMA in construction is helping the designers to make prefabricated building parts and components in such a way that their fabrication and assembly on site becomes easy and cost effective which will further increase the speed of construction (Ali et al., 2018).

Offsite construction is one of the emerging areas of construction industry in the world and will play a pivotal role in the development of country buildings infrastructure (Gan et al., 2018). The ability to harness the efficiency of manufacturing processes, offsite construction or prefabrication, has long been seen as the prospective future method to amplify production speed, quality and control of the construction sector (Pan & Goodier, 2011). Early collaboration is essential for any construction project and when it comes to Modular construction, that coordination becomes critical (Zolfagharian & Irizarry, 2017). It is fact that manufacturing industry has proved that it is more efficient than construction industry in production but there are only few attempts have been made to evaluate construction through design assessment (Abanda, F. H., J. H. M. Tah, 2019). Integrating construction knowledge into design phase has become critical, particularly considering the current demand for skilled labors in the commercial construction sector (Zolfagharian & Irizarry, 2017) . This study will provide early collaboration between stakeholders to optimize design and evaluation of all building elements (Manu et al., 2019). Resultantly product quality will improve, less material wastage will occur, more cost control, short project duration, less overlapping of activities etc. On average it is considered that modular construction is 50% quicker than conventional methods (Engineering, 2018). Pakistan is developing country where construction industry needs to modernize with BIM, where time and cost are key factors for decision making specially for mega building projects (Engineering, 2018).

BIM adoption in modular construction is getting its importance in many public and private funded projects of construction sectors in developing countries, but these sectors have limited use. The reason for such hesitation is non-familiarization with advantages of BIM tools to save cost, time, and material. Traditional prefabricated construction is not fulfilling the optimization needs and leads toward large construction material waste generation. Local industry of Pakistan can use this BIM based optimization tool as an “early-stage design capability” to create more effective design.

1.3 RESEARCH OBJECTIVES

- a. To identify design evaluation factors using following prefabricated design principles.
 - Design for Manufacture and Assembly (DFMA)
 - Lean construction principles.
- b. To formulate framework for BIM based design evaluation of prefabricated building elements from above derived factors.
- c. To integrate and validate the developed framework in BIM digital platform.

LITERATURE REVIEW

2.1 INTRODUCTION

Design efficiency is one of the key factors in prefabricated construction and design for manufacture and assembly (DFMA) is one of the leading techniques in manufacturing industry for design efficiency. This method is a design procedure and provide guidelines that supports design simplification through integration of economic aspects of materials and processes into the design with the goal of achieving efficient manufacturing and assembly of different building elements.

Offsite construction is one of the emerging areas of construction industry in the world and will play a very important role in the development of country buildings infrastructure (Gan et al., 2018). The ability to harness the efficiency of manufacturing processes, offsite construction or prefabrication , has long been seen as the prospective future method to amplify production speed, quality and control of the construction sector (Pan & Goodier, 2011). Early collaboration is essential for any construction project and when it comes to Modular construction, that coordination becomes critical (Zolfagharian & Irizarry, 2017). It is fact that manufacturing industry has proved that it is more efficient than construction industry in production but there are only few attempts have been made to optimize construction through design assessment (Abanda, F. H., J. H. M. Tah, 2019). Integrating construction knowledge into design phase has become critical, particularly considering the current demand for skilled labors in the commercial construction sector (Zolfagharian & Irizarry, 2017) . This study will provide early collaboration between stakeholders to optimize design and evaluation of all building elements (Manu et al., 2019). Resultantly product quality will improve, less material wastage will occur, more cost control, short project duration, less overlapping of activities etc. On average it is considered that modular construction is 50% quicker than conventional methods (Engineering, 2018). Pakistan is developing country where construction industry needs to modernize with BIM, where time and cost are

key factors for decision making specially for mega building projects (Engineering, 2018).

Design optimization and evaluation factors pointed out from literature below. From 25 papers and 5 books I have deduced the 121 design evaluation factors. **Error! Reference source not found.** below shows the information research paper and factors.

Table 1 Design Evaluation Factor from DFMA & Lean Principles

S. No	Design Evaluation and Optimization Factors from DFMA & Lean	Author
1	Duration to assemble/reduction in construction time	(Pan & Goodier, 2011) (Babalola et al., 2019) (van Vliet & van Luttervelt, 2004)
2	Size of the element	
3	Work force required	
4	Expertise Required	
5	Availability of element	
6	Standardization ability of element/ Standardize by utilizing normal parts, cycles, and strategies across all models and even across product offerings to allow the utilization of higher volume measures that ordinarily bring about lower item cost.	(Boothroyd, G., P. Dewhurst, 2001.) (Boothroyd, 1994)
7	Complexity of geometry	
8	Ease to handle	
9	Weight of Building elements/ Designers ought to take a stab at least weight predictable with strength and stiffness prerequisites	
10	Higher initial capital cost / Manufacturing cost	

11	Health and safety risk/ A good design idea from the safety standpoint is one whose manufacturing method does not involve any safety issue for the workers.	
12	Waste reduction	(A. R. Chini & Schultmann, 2002)(B. A. R. Chini & Bruening, 2008.)
13	Simplified design /Use the simplest cross-sectional shape	
14	Construction task independence	
15	Modularize ability	
16	Scaffolding needs for particular elements	
17	Accessibility of building element	(Quale et al., 2012) (Paez et al., 2005)
18	Adverse weather, construction capability of design	
19	How much fast we can assemble the building elements / simplify design for automation for automatic assembly line.	
20	Worker Training required	
21	Importance of conceptual stage design	
22	Appearance of element// How attractive the product is	
23	Recycle Ability of the Material of building element	
24	Ease to disassembly the element	
25	Client interest or benefit of prefabricated Building /Possible financial benefits of building element	(Webster & Costello, 2005)

26	Use mechanical connections rather than chemical ones / Avoid connections	
27	Number of fasteners or connectors should be minimum	(Yuan et al., 2018a)
28	Design the joints and connectors to withstand repeated use with different load condition	
29	Identification of component should be permanent, and barcode based.	
30	Use the structural grid for design	
31	Different types of components in building elements should always be minimum.	
32	The access of all parts and connection of building elements is essential.	
33	Save all information of the building components and materials and cost etc.	
34	End of life value of building element /How long the product lasts in use	
35	Reuse Ability of the building element	
36	Resale value of building element/material	(Steinhardt et al., 2020)
37	Environmental Hazard of Material/Design should be Environmentally Friendly / Weather resistance required of particular building element	
38	Layer building system	
39	Salvage value of material at end of life	

40	Services (plumbing, electrical, HVAC) separation from structure/serviceability requirement of element / operational cost	
41	Durable label of Material	
42	Score of Manufacturing ability during design phase	
43	Design coordination mechanism existence	(Howard & Lewis, 2003)
44	Dependencies between design activities	
45	Modelling of tolerance and surface roughness	
46	Symmetry of design	
47	Modelling of tolerance and surface roughness requirement	(Connor et al., 1987)
48	Integration of design teams members	
49	Customer requirement	
50	Market demand// Short time-to-market	(Das & Kanchanapiboon, 2011)
51	Legislation	
52	Basic knowledge of engineering i.e., material, manufacturing physics	
53	Function of product, operation action and maintenance	
54	Methods and technique required to manufacture the product	(Bogue, 2012)
55	Selection of material and product design	
56	Historic design information	

57	Knowledge about possibilities of manufacturing, outsource options	(Connor et al., 1987)
58	Knowledge regarding standards	
59	Dimensions required	
60	Tolerance tightness level and surface roughness required	
61	Information regarding machine capabilities and processing	
62	Design appropriate to the expected level of production	(Li et al., 2020)
63	Ease of fabrication	(van Vliet & van Luttervelt, 2004)
64	design such that, Number of parts are minimum	(Mubarak, 2001.)
65	Design according to the capabilities of manufacturing process	
66	Avoiding process restrictiveness. /Reduce the Manual Handling Time /For manual assembly, both access and vision are not restricted.	
67	Design in such a way that general-purpose tooling/Process should be used.	
68	Sharp corners should be minimum; use fillets and radii / Introduction of guides and chamfers in design to reduce assembly time	(Boothroyd, 1994)
69	Repositioning is time consuming so design should use minimum repositioning for its assembling	

70	Space between holes of building element should be such that it is easy to connect with another element	
71	The need of central design database / data base is needed for assembly times and cost factors for various design situations and production conditions storage.	
72	Reliability of component	(Belanche et al., 2021)
73	Cost data of different materials for different building components	
74	Avoid designs with inherently high scrap	(Tauriainen et al., 2016) (Manu et al., 2019)
75	rates.	
76	Welding is required or not	
77	Which standard shape is available and easy to make for particular building element	
78	Metal stamping process requirements for particular elements	
79	Suitable materials for stampings	
80	Heat conductivity of material required for particular building elements	(Ekanayake & Ofori, 2004)
81	Heat capacity of material required for particular building elements	
82	Durability of material required for particular building elements	

83	Self-tapping screws for window design of building is much better option instead of using nuts or threaded holes in mating of parts.	
84	Cast or molded identification is better than labels.	
85	Use sub-assemblies	(Goudeau, 2015)
86	Design parts so that they cannot be inserted incorrectly	
87	Use layered, top-down assembly design approach// Use pyramid assembly—provide for progressive assembly about one axis of reference	
88	Avoid designs that require the simultaneous addition of several parts that must be kept in alignment as they are added to the main assembly	
89	Avoid assembly designs that require parts to be manually held in place until other parts are inserted because this kind of situation has some risk of quality problems as well as additional assembly time and cost. // Long-Term Quality of design	
90	Welded assemblies should be made up of as few parts as possible.	(Bralla et al., 1996)
91	Weld joints should be placed so that there is room for easy access of the welding nozzle.	(Connor et al., 1987)

92	Assembly should be designed so that the welded joint is horizontal	(Li et al., 2020)
93	Welds are avoided where appearance is most important	
94	The joint should be designed so that it requires minimal edge preparation	
95	Welding operation should be performed on subassemblies rather than major complex assemblies	
96	Smooth surfaces are preferred. Smooth surfaces are more easily wet by a spreading liquid adhesive (Windows)/ The type of surface texture customer needs	
97	Design for Function and Performance specific	
98	Is the product easy for the user to install and operate? / User friendliness or ergonomics	
99	Ship ability of design	(van Vliet & van Luttervelt, 2004)
100	Upgradeability provisions of design	
101	Provide features that prevent jamming of parts	
102	Design so that there is little or no resistance to insertion and provide chamfers	
103	The necessity for holding parts down to maintain their orientation during manipulation of the Subassembly or during the placement of another part 3.9 fig	

104	Design so that a part is located before it is released. FIG 3.10	
105	Parts Requiring Two Hands for Manipulation	
106	Effect of Assembly layout on Acquisition Times	
107	Tangling and Nesting provisions for different building elements for better handling during assembly	
108	If the degree of a part's deformations as a result of the applied loads is crucial, then the resistance of material to deformation will be a factor.	
109	In-Service Temperature / roof, walls	(Das & Kanchanapiboon, 2011)
110	Exposure to Moisture (Fresh Water, Sea Water) / roof, walls, floor / Provide room for drainage of water/Fluids	
111	Transparency and Colour Requirements / interior design of kitchen and bathroom	
112	Use of snap fits	
113	User feedback of particular design	
114	Design the product so that parts prone to wear or failure are easily visible and accessible	
115	Reduced amount of storage in the construction site	

The design evaluation factors from design for manufacture and assembly (DFMA) and lean construction principles (LCP) are separated. The Table 2 below shows the factors of design optimization and evaluation those was scrutinized and analysed for their literature score and following results were obtained.

Table 2 Field Survey of Design Evaluation Factors from Lean Principles

S. No	Design Evaluation factors from Lean Principles	Sum	Average	RII	Rank
1	Dependencies between design activities.	140	4.117647059	0.823529412	1
2	Retaining all details of building components (strength, material, cost, dimensions, etc)	138	4.058823529	0.811764706	2
3	Time required for assembling of the building element	135	3.970588235	0.794117647	3
4	Access and vision restrictions for Manual assembly	135	3.970588235	0.794117647	3
5	Modularize ability of the building element	134	3.941176471	0.788235294	4
6	Size of the building element	133	3.911764706	0.782352941	5
7	Number of different components in building elements	130	3.823529412	0.764705882	6
8	ship-ability	130	3.823529412	0.764705882	6
9	Ease of fabrication	129	3.794117647	0.758823529	7

10	Symmetry of design	128	3.764705882	0.752941176	8
11	Less waste generation	126	3.705882353	0.741176471	9
12	Designer Knowledge about possibilities of manufacturing, outsource options	126	3.705882353	0.741176471	9
13	Function and Performance requirement of the particular Building element	125	3.676470588	0.735294118	10
14	Amount of storage required at the construction site.	125	3.676470588	0.735294118	10
15	The need for High-Speed Automatic Assembly and Robot Assembly	123	3.617647059	0.723529412	10
16	Reuse Ability of building component	123	3.617647059	0.723529412	10
17	Easy handling of Building Element	122	3.588235294	0.717647059	11
18	The requirement of simultaneous addition of several parts that must be kept in alignment	122	3.588235294	0.717647059	11
19	Coordination mechanism	121	3.558823529	0.711764706	12
20	Tangling and Nesting provisions for different building	121	3.558823529	0.711764706	12

	elements for better handling during assembly				
21	Applied loads and resistance of the material.	121	3.558823529	0.711764706	12
22	The layered (top-down assembly) design approach	120	3.529411765	0.705882353	13
23	Specialized workforce requirement	120	3.529411765	0.705882353	13
24	Effect of assembly layout on Acquisition Times	119	3.5	0.7	14
25	Availability of building element as per need	119	3.5	0.7	14
26	Market demand	114	3.352941176	0.670588235	15
27	Use of moulded-in identification instead of attached labels.	111	3.264705882	0.652941176	16
28	Expected Level of Production	107	3.147058824	0.629411765	17
29	Salvage value of Building component	105	3.088235294	0.617647059	18

Design Evaluation factors from DFMA and their analysis based on initial field survey from experts as shown in the Table 3. This field survey was conducted to actually rank the importance of already find factor towards design evaluation of prefabricated building elements.

Table 3 Field Survey of Design Evaluation Factors from DFMA

S. No	Factors from DFMA for Design optimization and Evaluation	SUM	AVG	RII	Rank
1	Complexity of geometry	146	4.294118	0.858824	1
2	Environmental Friendliness	139	4.088235	0.817647	2
3	Health and safety risk involved in design option	138	4.058824	0.811765	3
4	Services (plumbing, electrical, HVAC) separation from structure or serviceability requirement of design	138	4.058824	0.811765	4
5	Importance of conceptual stage design	137	4.029412	0.805882	5
6	Design performance for exposure to Moisture (Fresh Water, Sea Water) and drainage	137	4.029412	0.805882	6
7	Higher initial capital cost / Manufacturing cost	136	4	0.8	7
8	Simplified design /Use the simplest cross-sectional shape	135	3.970588	0.794118	8
9	Selection of material and product design relation	135	3.970588	0.794118	9
10	Manufacturing ability of the existing assembly line.	135	3.970588	0.794118	10
11	Heat conductivity of material required for particular building elements	134	3.941176	0.788235	11
12	Expertise Required to design particular element	133	3.911765	0.782353	12
13	Tolerance tightness level and surface roughness requirement	132	3.882353	0.776471	13

14	Ease of disassembly the building element	132	3.882353	0.776471	14
15	The necessity for holding parts down to maintain their orientation during manipulation of the sub-assembly.	132	3.882353	0.776471	15
16	Basic knowledge of engineering requirement of particular design i.e., material, manufacturing physics	131	3.852941	0.770588	16
17	Does Information regarding machine capabilities and processing need at the initial design stage matter?	131	3.852941	0.770588	17
18	Use sub-assemblies	131	3.852941	0.770588	18
19	Cost data for different materials for different building components.	130	3.823529	0.764706	19
20	The durability of material required for particular building elements design	129	3.794118	0.758824	20
21	Visibility and accessibility of parts prone to wear or failure	129	3.794118	0.758824	21
22	Recycle Ability of the Material	129	3.794118	0.758824	22
23	Client interest or possible financial benefit of design	128	3.764706	0.752941	23
24	Function of building element, operational action, and maintenance requirement	128	3.764706	0.752941	24
25	Customer requirement importance	128	3.764706	0.752941	25

26	Reliability of individual components to make the design more reliable.	128	3.764706	0.752941	26
27	Heat capacity of material required for particular building elements	128	3.764706	0.752941	27
28	Bye law's requirement of design from development authorities	127	3.735294	0.747059	28
29	The ability to use general-purpose tooling and process	127	3.735294	0.747059	29
30	Design compliance with standards	126	3.705882	0.741176	30
31	Dimensions requirement	126	3.705882	0.741176	31
32	Standardization ability of Building element design	125	3.676471	0.735294	32
33	Design apart so that as many manufacturing operations as possible can be performed without repositioning it.	125	3.676471	0.735294	33
34	Upgrade ability provisions of design	125	3.676471	0.735294	34
35	Number of parts in Welded assemblies	124	3.647059	0.729412	35
36	Methods and technique required to manufacture the product	124	3.647059	0.729412	36
37	In-Service Temperature requirement of particular building element	124	3.647059	0.729412	37
38	Designing parts such that they cannot be inserted incorrectly or zero fitting problem	123	3.617647	0.723529	38
39	The type of surface texture customer needs	123	3.617647	0.723529	39

40	which standard shape is available and easy to make for particular building element	123	3.617647	0.723529	40
41	Appearance of element or how attractive the design of product.	122	3.588235	0.717647	41
42	Jamming of parts during installation	121	3.558824	0.711765	42
43	Scaffolding needs for the particular design of elements	120	3.529412	0.705882	43
44	Self-tapping screws use for particular building element	120	3.529412	0.705882	44
45	Design of the product having part is located before it is released.	120	3.529412	0.705882	45
46	Manual Handling requirement of design.	119	3.5	0.7	46
47	Transparency and Colour Requirements for kitchen and bathroom	119	3.5	0.7	47
48	User-friendliness or ergonomics of the design	119	3.5	0.7	48
49	Use of snap fits	119	3.5	0.7	49
50	Use of the number of fasteners or connectors	118	3.470588	0.694118	50
51	Scrap rates of design	118	3.470588	0.694118	51
52	Appropriate space between connecting holes.	117	3.441176	0.688235	52
53	Weight consistent with strength and stiffness requirements	116	3.411765	0.682353	53
54	Edge preparation required for a particular building element.	116	3.411765	0.682353	54
55	Use a structural grid base design	115	3.382353	0.676471	55

56	Use mechanical connections rather than chemical ones	114	3.352941	0.670588	56
57	Score of Manufacturing ability during design phase	114	3.352941	0.670588	57
58	The importance of Historic design information	114	3.352941	0.670588	58
59	Amount of resistance to insertion and requirement of chamfer	112	3.294118	0.658824	59
60	Designing of building element in which Parts Requiring two hands for Manipulation	112	3.294118	0.658824	60
61	Establishing a database that consists of assembly times and cost factors for various design situations and production conditions.	111	3.264706	0.652941	61
62	Metal stamping process requirements for particular elements	110	3.235294	0.647059	62
63	Suitable material for stamping	103	3.029412	0.605882	63

A preliminary survey was conducted among the professionals working in the field related to prefabricated construction industry. The survey questionnaire asked about relevancy of the design evaluation and optimization factors derived from lean principle and design for manufacture and assembly (DFMA). A survey form was made on google forms and circulated via links on email and other social media facilities. Due to unavailability of proper data a sample size of 30 was assumed and in total 36 responses were recorded. Following table shows the demographics of the students who completed the questionnaire.

Table 4 Field Survey, Experts Experience Distribution

Experience	No of Questionnaires filled	Percentage
less than 5 years	2	6.060606
6-10 years	7	21.21212
More than 10 years	24	72.72727

The results of the preliminary survey indicate that these are the most important design evaluation and optimization factors for prefabricated building elements. These factors can be used to make BIM tool to evaluate the design at initial stage. Table 4 represents the experience of industry experts.

To make BIM based design optimizer we first need to dig out more about lean and DFMA guidelines to understand how these factors will be used for design optimization and evaluation at initial stage of project.

2.2 DESIGN OPTIMIZATION APPROACH IN CONSTRUCTION INDUSTRY

2.2.1. Design of experiments approach

Building simulation-based optimization includes direct coupling of the optimization calculation to a simulation model, making it computationally concentrated. To beat this issue, a methodology is proposed utilizing a mix of trial design strategies (fragmentary factorial design and reaction surface technique) (Dhariwal & Banerjee, 2017). These strategies rough the simulation model conduct utilizing proxy models, which are a few significant degrees quicker than the simulation model. Fragmentary factorial design is utilized to distinguish the huge design factors. Reaction surface approach is utilized to make substitute models for the yearly cooling and lighting energy with the screened huge factors. The mistake for these models is under 10%, approving their viability. These substitute models accelerate optimization with hereditary calculations, for sing leand multi-target optimization issues and situation investigations, bringing about a superior

arrangement. Hence, optimization becomes conceivable inside sensible computational time with the proposed procedure (Huber, 2011).

The design evaluation of prefabricated building elements can be enhanced by using multiagent system for multidisciplinary field such as architectural design modification, building engineering and facility management (Yin et al., 2019). This method is very effective as it is using design inputs and evaluation methods in virtual environment. This method algorithm will use real time data for calculation. An evaluation method has implemented various optimization and evaluation algorithms and also be incorporated with building simulation software's.

(Dhariwal & Banerjee, 2017) has stated that some simulation models offer fast fitness function evaluation of prefabricated buildings algorithms and their limitations. This shows that if we implement building simulation model combined with GA to evaluate the fitness function of building, this would be very intensive as for as computation is concerned. The use of more efficient computation system can increase the GA ability to resolve the real-world problem instead of exact evaluation of the system. This is possible because efficient algorithm of computation will easy implementation of GA system for the evaluation of data of prefabricated building element.

2.2.2. Grey situation decision-making model

World energy need tremendously growing with time and construction sector is one of the major energies consumers. In future green building concept will get most attraction for energy conservation and their ability for environmental protection. The green building architecture has two major layers: first part consists of design presentation and business potential of design, and other part is the database for the previous part for analysis. The grey situation decision making model is used to evaluate and optimize the whole database information to get the most efficient outcome. The application result of the said method has proved that it can reduce the construction cost significantly by improving quality of the design of prefabricated building elements (Zolfagharian & Irizarry, 2017).

2.2.3 Meta-heuristic optimization algorithms

The experts believe that the world will have newborn 4 billion additional people in 60 years, it means that approximately 1,200,000 – 1,500,000 residential units will be required annually to fulfil the burgeoning demand, and this herculean task cannot be achieved through traditional building construction methods.

This creation can be filled by utilizing pre-assembled structures in the development business, a high-level technique that enjoys different benefits like diminishing the structure cycle time, simple creation, cost decrease, and top-notch control. On the opposite side, the lack of crude material and mechanical progressions propel existing social orders to search for streamlined arrangements in their ways of life and businesses. It is likewise pertinent to building ventures that have an assortment of targets, for example, least time and cost alongside accomplishing more dependable structures. These include subtleties of the constructions and development industry from the configuration stage to the site task plan stage. A primary objective of advancement in precast constructions is to track down the ideal area of the components which is characterized as the 'geography improvement'. Geography streamlining has a wide scope of uses is supported designs fully intent on decreasing the expense of constructions. Utilization of Artificial Intelligence (AI) under the fulfillment of configuration rules and norms is an appealing way to deal with accomplishing this reason (Baghdadi et al., 2020).

Artificial intelligence (AI) and fast speed of data communication has made a way of real time collaboration between engineers and architects. This early and time-based collaboration will enhance the project execution. This method provides the easy application of Artificial intelligence (AI) in the prefabricated construction industry. Currently structural engineering has not used the Artificial Intelligence (AI) in the conventional construction industry. Despite of many benefits AI has failed to get implemented in the true spirit to improve the process of prefabricated construction. MATLAB and AI based design evaluation algorithm with Particle Swarm Optimization (PSO) if combined , can provide design analysis and evaluation at the conceiving stage of the construction project.(Baghdadi et al., 2020).

2.2.4 Genetic Algorithm

Recent research shows that a multi objective design optimization and evaluation of prefabricated building is possible by means of GA, for example work of Ascione, Mauro and others. They have used wide range of multi propose energy optimization and evaluation of buildings. Some researchers including Khairi, Jafari and Valentin has shown particular interest in the field of objective functions for design optimization. I have studied the that until now, the most deep and complete review of selection design optimization and evaluation criteria has been made from various studies on residential construction sector by Harlequin framework. China and Italy researchers mainly focusing on residential construction design modification. Open access software and GA algorithm combinedly employed by many researchers and they have consider it very effective tool for design optimization and evaluation of residential buildings (Hyun et al., 2021).

The national and international organizations have encouraged to reduce the carbon emissions of building and has applied restrictions, fines to implement their regulations. So due to this reason designers must consider the new restriction and regulations of respective organization while designing the prefabricated building elements to make design according to sustainable construction. As we know that passive and active guidelines for sustainable construction have not applied at initial stage of design. On the basis of above-mentioned lack of centralized system, we can say that a new approach to achieve multi-objectives of design evaluation of new buildings elements like building envelop, wall, roof and kitchen as a whole is needed. This will allow modification of architectural variability (Ciardiello et al., 2020).

2.3 DESIGN FOR MANUFACTURE AND ASSEMBLY (DFMA)

The technique of design for manufacture and assembly (DFMA) approach has given tremendous running advantages to industry in outfitting item design which are basic and conservative to create. The technique has produced strong competitive advantages to the companies that have used it. However, low cost in manufacturing is only one objective of a sound product design. There are many other desirable objectives-quality of product being one of them.it is very obvious that those experts that are using DFMA can use it in broader way in construction industry. Due to the

implementation of this method desired goals of good product design can be achieved. This technique has successfully proved its importance for design modification in manufacturing industry. The following objectives can be achieved very easily like user friendliness, safety of design, serviceability, sustainability requirement of modern age construction, economic viability of design, ability of the design to market in very short time, quality of the design. The goal of this book, then, is to explain how the DFM technique is evolving and how it must evolve into an approach that more strongly and more specifically addresses the broad series of important objectives of sound product design. They can be served through the same knowledge-based technique that works so effectively in improving manufacturability. The latest available method design for X number functions (DFX), where X indicates what are those factors that need to be improved in design at initial stage of design of any project (Boothroyd, 1994). The DFMA guidelines and their benefits are shown in Table 5.

Table 5 DFMA Guidelines

Sr. No	Guidelines	Benefits
1	Minimize part count	It will reduce time of assembly
2	Use Standard, off the shelf parts as compared to customized components.	Reduce cost, increase production speed
3	Minimize and standardize the number of components.	Less cost of components, improve reliability
4	If possible, use same material for all components	Cost reduction and material reuse
5	Use minimum different parts	It will increase assembling time
6	Design for easy fabrication	It will accelerate the fabrication speed
7	Consider modular design options	It will help to generalize the design and automate the whole assembly line

8	Aim to remove mistakes at initial stage	It will remove on site remodification and hence time to assembly and delivery will be reduce
9	Design for best handling and orientation	Handling will consume a lot of time .it will reduce handing time and machine selection for handling is previously non
10	Design according to the capacity of assembly line.	Assembly line capabilities should be known and hence time of fabrication will reduce
11	Design for automated assembly fabrication	Design that can be easily automated will increase production rate to meet customer needs

Design for manufacturability (DFM) can have two definitions. first definition tells us that DFM consist of step for design modification at initial stage. This modified design is easy to manufacture, and less manufacturing cost will occur. In a more specific way, we can say that as per the definition of the book that DFM is a knowledge-based method which provides us multiple guidelines and principles, experienced based easy to manufacture methods so that it would be easier to make the product.

These guidelines and recommendations may help in many common product features, but their primary task is to enhance the manufacturability (Yuan et al., 2018b). The manufacturability means the ease and comfort by which a product can be produced, such as, minimization of labour, easy configuration, reduced labour costs, material costs and overhead costs. All these factors are manifested as a lower manufacturing cost. Why design for manufacturability rather than for function, quality, reliability, etc.? Because, historically, manufacturability has been overlooked, primarily in favour of designing for features, function, and appearance.

In fact, there is a gold mine of cost benefits to be tapped if manufacturability is addressed when a product is designed.

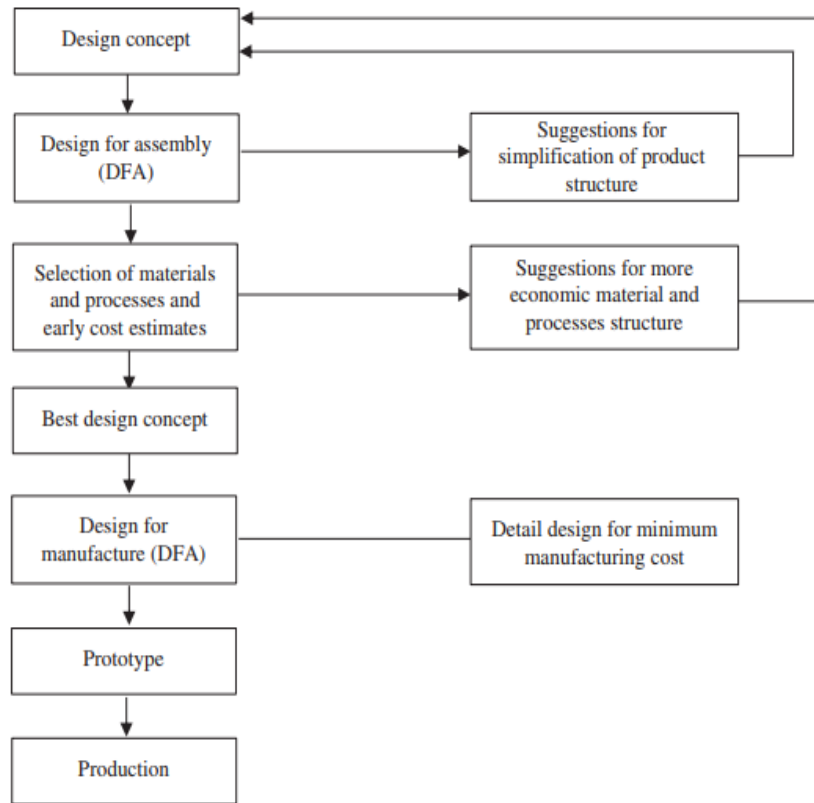


Figure 1 DFMA Flow chart

2.4 LEAN CONSTRUCTION PRINCIPLES

According to (Jin et al., 2018), the concept of Lean construction continues to evolve due to which it has no specific definition. co-founders of Lean Construction Institute (LCI), Howell and Glenn Ballard consider lean principle as the method of construction management (Gao et al., 2018). The lean construction principles provide the basis of project delivery process. Lean construction provides operational based project management like design-build. The new method tremendously improved the assembly management of prefabricated building design in factory environment (Gao et al., 2020).

Ford's operation studied by Toyota, and they have found many possibilities for improving the design auto-production plant in Japan. Chief engineer of Toyota has visited the site many times. Both of the stakeholder has realized that there was so much waste in Ford's operation, only assembly line workers putting some value for the improvement operations of assembly line. Design experts are only responsible for production process designing .they are also responsible for creating the guidelines for workers of assembly line and assembly line in charge is ensuring the compliance of rules (Forbes & Ahmed, 2010). The team has made ring of important lean principles as shown in Figure 2. The team of industrial engineers have proposed the seven-waste reduction system (Forbes & Ahmed, 2010).

- Overproducing
- Idle time waste
- Transporting/conveyance waste
- Processing waste: waste in the work itself
- Inventory waste (having unnecessary stock on hand)
- Wasted operator motion (using unnecessary motion)
- Producing defective goods (waste of rejected production)

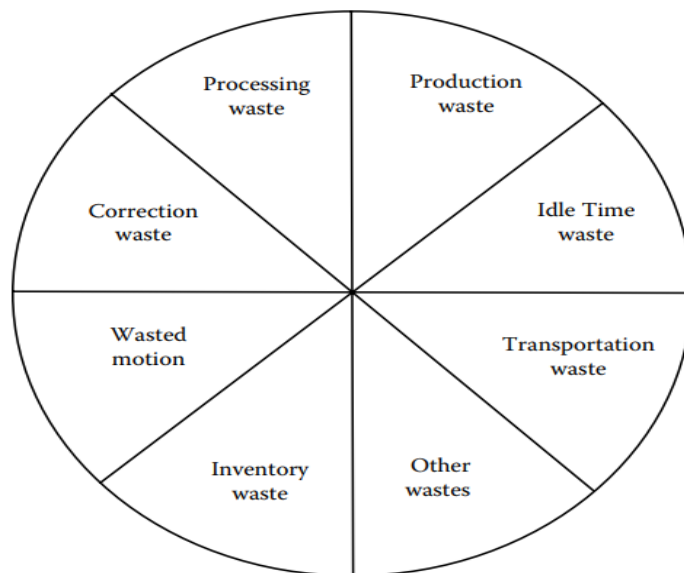


Figure 2 Lean construction principles

2.4.1 Origins of Lean Construction Method

The research work by Frank Gilbreth in the manufacturing industry has identified the potential of lean principles for the improvement of construction industry. He is known as father of industrial engineering as he later built on Frederick Taylor's principles.

The Gilbreth noted the process of bricklaying and outlined that the process can be improved in efficiency. He proposed the idea of "speed work" and said that the various motions of bricklayers were unnecessary and time-consuming. He has proposed the method of fast brick laying and time has reduce to much extent. He also prepares the guild lines for works performance improvement for assembling operation for prefabricated building element. It will increase the productivity rate to a great extent. Lean construction principle, thus saving them from the construction responsibility. The improvement in the productivity growth mainly due to the application of lean principle technology (Babalola et al., 2019).

Lean Project Delivery and Integrated Practices

The Gilbreth has proposed the following main guidelines for lean construction which has tremendously improved the construction operations the types of lean methods are also shown in Figure 3.

- Variability reduction in design
- Cycle time reduction of one unit production
- The number of steps should be minimum, a smaller number of parts, and linkages.
- The output flexibility should me more.
- The whole production process should be transparency.
- The automated control system should be implemented on the complete process.
- Research and development should continuously improve the process.
- Benchmark should be setup to monitoring the whole process.

Some points are very important in relation with the heuristic principles of design evaluation of prefabricated building elements are:

- Those value should be highlighted which are adding no value for design improvement.
- The supplier and customer for each activity should be previously known.
- The standard procedures adoption should be streamline and the whole flow process transparent.
- The entire flow process should be optimized.

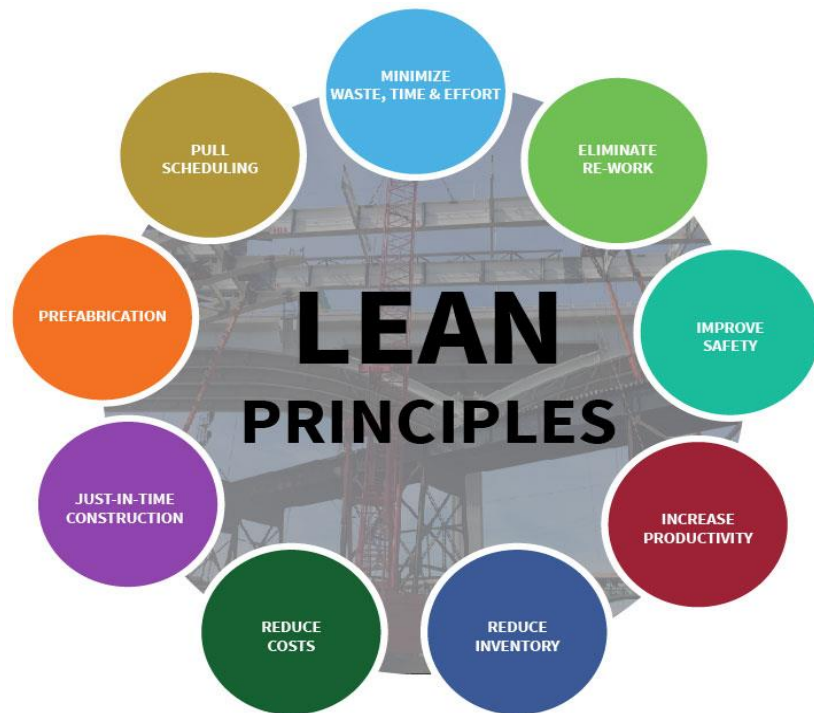


Figure 3 Types of lean construction Principles

2.4.2 Lean Design and Construction

According to (Aziz & Hafez, 2013) Lean construction principles are consists of the applications lean technique to get the benefits for processes improvement of whole construction process of prefabricated building design.

Benefits include:

- Less cost will occur
- The minimum number of delays
- The uncertainty in the whole construction process should be minimum
- Waste minimization of prefabricated building design

- The green building and energy efficiency of design should be considered
- Design should be according to the customer needs.

Lean principles application has resulted the better use of all resources for example, material, cost, and labor. It has also improved the quality of the construction of prefabricated building element, the safety aspects of construction and above all the satisfaction of all stakeholders such as builders, clients, and design experts. All construction projects are unique in nature and complex but lean construction principles of production management has made very important improvement specially for prefabricated building elements.

Lean construction technique has been used on many projects successfully for example even in small office building it has reduced the cost of construction by 25 % and time of construction 12 to 3 weeks , which shows the level of potential that lean principles have for construction process improvement (Jin et al., 2018).

2.4.3 Philosophical Differences between Lean Construction and Traditional Construction

Lean construction management is entirely different from the traditional construction management. The method has clear workflow system and planning procedure to accelerate and improve the project outcomes. The same technique can also be applied for prefabricated building design evaluation and improvement (Goh & Goh, 2019).

In conventional construction practices individual crew or labor will be rewarded on completion of task while in lean construction the work is divided into sections and upon completion of section, all involved labor and individual will be rewarded. The sense of reward is so centralized and resultantly the productivity of the whole process accelerates exponentially. This can only be achieved when all stakeholders such as professionals, design experts, contractors and project managers are connected with centralized communication channel for example BIM infra of Autodesk.

Lean construction philosophy suggested that one should build the basic infrastructure and institute at or off site of the project to manage the whole

construction and assembling operations. It also includes the efficient workflow between the construction team members. There is no interruption between the communication of construction information. Therefore, there is excellent communication between crews and overall everyone is focusing on his job to complete the task efficiently by showing his self-interest in the task (Connor et al., 1987). Lean construction technique has not replaced the CPM or other methods, but it works within the CPM domain with some improvements.

In CPM we have to identify the major project milestones and the sequencing of the most critical activities but in lean construction we have to do the same for short term like five weeks planning to achieve the tasks. It is very difficult that using CPM we can track or plan the individual task for short duration. But using lean principles we can do it very effectively.

Lean construction emphasizes on the planning which is very important as mentioned in the PMBOK and it can optimize the whole planning very efficiently. The cutting-edge technology of manufacturing industry should be introduced for construction, and it is greatly encouraged by lean principles. In offsite construction we need continuous production of construction units. Last planner system technique is best because it focuses on the short interval of time so it will reduce the cycle time of production. Lean construction emphasizes on the value addition of product as compared to just controlling just time and cost (Forbes & Ahmed, 2010).

2.5 LEAN DESIGN MANAGEMENT

The purpose of lean design is rightly defined by the British Institute of Architectural Technologists (BIAT): “the process of improvement of manufacturing ability of the design is to be done through effective communication of concerned information”. The lean Design management technique should be following the lean principles for construction as suggested by the BIAT.

Some of the suggested principles are the following:

- The complete understanding of value addition in product
- Customer point of view should be identified at start.
- The waste minimized flow process should be developed

- The required information of any step should be easily available.
- Construction design improvement cycle should be effective to achieve the most efficient design.

Some tools and techniques recommended for lean design are:

- Low level team member involvement in design improvement and the all-team members in better way for best coordination.
- Generative design approach should be followed, and simulation of all possible option should consider.
- Scheduling and design matrix should be planned accordingly.
- First utilize the production of small sample batch and evaluate the bath on already set principles
- Use latest technology that support the use of lean design such as BIM.

2.6 LAST PLANNER SYSTEM

The last planner system effective implementation is so critical. The construction project is complex in nature and smooth workflow is difficult but using last planner system material resource can be made more productive. In conventional construction management the whole working schedule is not in line with all workers. recent research shows that using conventional construction there are only one-half tasks of assembling a product that given to works are likely to be completed in the given schedule. Ballard and other authors have published the study which shows that how workflow plan of various projects have failed. Its failure percentage is 54 (Babalola et al., 2019).

Workflow of different project is very complex which importantly influence the lean construction practices. The delay in one activity will leads to delay the whole project. The LPS use the lean construction guidelines to provide better control of the project. Planning the whole schedule of project is not effective till the time it is not linked with the task specific person in the manpower. Last planner system provides best use of material and labor resource. The Ballard has further refined the use of Last planner system to improve the performance offsite construction and has created

the centralized information sharing for crew which empowers them to complete their task effectively.

The Last Planner System Principles for improvement the assembling operation of Modular construction:

The forecast of construction project workflow should be realistic.

- Detail work plan is required for worker to complete task.
- Labour resource assignment should be consider while making plane
- Effective teamwork will identify the constraints and must be removed
- Worker or whole construction team commitments must be realistic
- If planning will fail it must be treated, consider as leaning opportunity, and never be considered for negative action

The last planner system divided into 4 level of project schedule and project planning. Different projects whose speed is fast like EPC projects involve complex engineering issues which may need working scheduling based on daily working.

- a. The Pull master scheduling
- b. The look-ahead scheduling of Prefabricated Building construction
- c. Weekly working plane of project (WWP)
- d. Daily working plan for Crew

2.6.1 The Pull master scheduling of construction project

Master scheduling shows the complete information of all important phases of the project and related documentation about the milestones of the whole project. In conventional construction practices master scheduling plan is prepared by the contractor which should be considered by the client. Major stages of the project have mentioned in the master scheduling (Gan et al., 2018).

2.6.2 The look-ahead scheduling of Prefabricated Building construction

The phase plan of master scheduling is used to drive time-based phases plan of whole project of 8-10 weeks of all the construction tasks. The same schedule will be used for workflow creation. The look ahead scheduling will also be used for pulling all activities from the master schedule.

2.6.3 Weekly Working Plan for crew (WWP)

The weekly working plane for crew will be drive from the above-mentioned look-ahead schedule. The detail work plan of each phase can be made. In every week schedule will be updated based on last planner system.

2.6.4 Daily working plan for Crew

Daily working plan for crew like engineer, procedure will be drive from weekly working plan .it will provide best accurate control then weekly working plan.

RESEARCH METHODOLOGY

3.1 INTRODUCTION

This Research methodology chapter describe the pathway for achieving the objectives described in chapter 1. The research design is disclosed first followed by background of data collection of design evaluation factors for prefabricated building elements and detailed research process including the framework development and its implementation.

3.2 RESEARCH DESIGN

For the first objective of our research, a detailed literature review has been executed from research papers to identify research gap and further the in-design optimization end evaluation factors from DFMA and Lean principles that exist in our current design optimization methods.

After this, literature analysis will be done to find out the most important factors to achieve design optimization and evaluation of prefabricated building elements. A preliminary survey has also been carried out among the professionals in our construction industry, to validate and reduce the number of factors for design optimization and evaluation of modular construction identified from literature.

In order to achieve the second objective, we studied the potential of BIM and design for manufacture and assembly (DFMA) and lean principles discussed in the literature to eradicate the design optimization problem at the start of project.the whole research method is explained in pictorial view as shown in Figure 4.

The shortlisted factors on the bases of industry input will be used for design optimization and evaluation of different prefabricated building elements.

Furthermore, a Design optimization and evaluation plugin will be created in Revit based on framework developed by using DFMA and lean factors. Through this centralized BIM based evaluator will be formed to optimize and evaluate the design at starts of project.

In the last phase Validation of this application will be done by communicating it to industry professionals. And the results will be analysed using SPSS or weighted average index method. Following chart shows the workflow of this research.

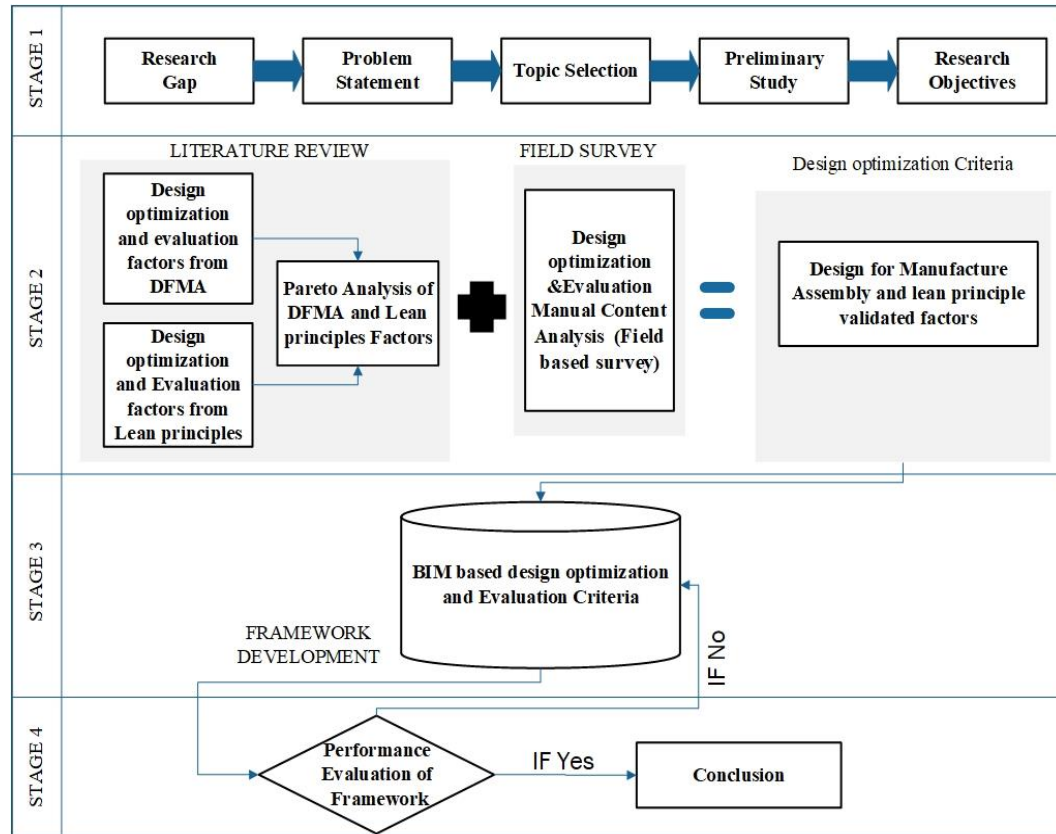


Figure 4 Research Methodology

3.3 IDENTIFICATION OF DESIGN EVALUATION FACTORS FOR PREFABRICATED BUILDING ELEMENTS FROM DFMA

In the first phase design evaluation factors for prefabricated building elements have been identified through substantial literature review.

After exhausting the said literature, 31 papers and 5 books were consulted to obtain a total of 80 design evaluation factors. Owing to overlapping problems with different names and same meaning were reviewed, content analysis was also performed to screen out less significant factors. Figure 3.3 shows similarity-based factor reduction. It led to shortlisting of 63 design evaluation factors for prefabricated different building elements.

3.4 IDENTIFICATION OF DESIGN EVALUATION FACTORS FOR PREFABRICATED BUILDING ELEMENTS FROM LEAN PRINCIPLES

In the first phase design evaluation factors for prefabricated building elements have been identified through substantial literature review.

After exhausting the said literature, 15 papers and 2 books were consulted to obtain a total of 45 design evaluation factors. Owing to overlapping problems with different names and same meaning were reviewed, content analysis was also performed to screen out less significant factors. Figure 3.4 shows similarity-based factor reduction.

It led to shortlisting of 29 design evaluation factors for prefabricated different building elements.

3.5 PRELIMINARY STUDY FOR VALIDATION OF DESIGN EVALUATION FACTORS FROM DFMA & LEAN PRINCIPLES

After extensive literature review and similarity-based content analysis 92 factors have been considered. A survey form was developed to get experts input to further scrutinize the design evaluation factors from DFMA and lean principles. The preliminary input was taken on Likert scale.

The initial survey was designed to take the input about the significance of factors towards design evaluation of prefabricated building elements. Experts was informed that these factors will be used to evaluate the design of different building elements at the initial design stage. The Figure 5 below is providing the information that is asked from experts.

Rate the significance of the following factors according to the Likert scale. The Likert scale is as follows.

- Not significant
- Slightly significant
- Significant
- More significant

- Extremely significant

3. Less waste generation *

Less waste generation: The design which generate less waste martial during and after construction. Waste minimization is any technique that either avoids, eliminates or reduces waste at its source.

Mark only one oval.

1	2	3	4	5
☐	☐	☐	☐	☐

Figure 5 Sample of Field survey form of Likert Scale

3.6 DEVELOPMENT OF FRAMEWORK FOR BIM BASED DESIGN EVALUATION

The framework is formulated in a way to add design evaluation-based system as integral part of BIM. As per the formulation the system is to be designed in such a way that it can evaluate the design of prefabricated building element at initial stage.

Figure.1, shows the whole conceptual model for BIM integrated design evaluator based on DFMA and lean principles.

The same method was used for lean principles as well.

3.7 DEVELOPMENT OF PLUGIN BIM-DE4PB

In the second phase, prototype is developed to evaluate the design of prefabricated building element. In Figure 6 whole framework of plugin development explained.

Also, in this phase of research, Building Information Modelling software API functioning is studied, and a plugin was developed for BIM based design evaluator for prefabricated building elements (BIM-DE4PB) for efficient design at initial stage of the project. To extend functionalities of most of the BIM softwares API is provided (Manu et al., 2019).

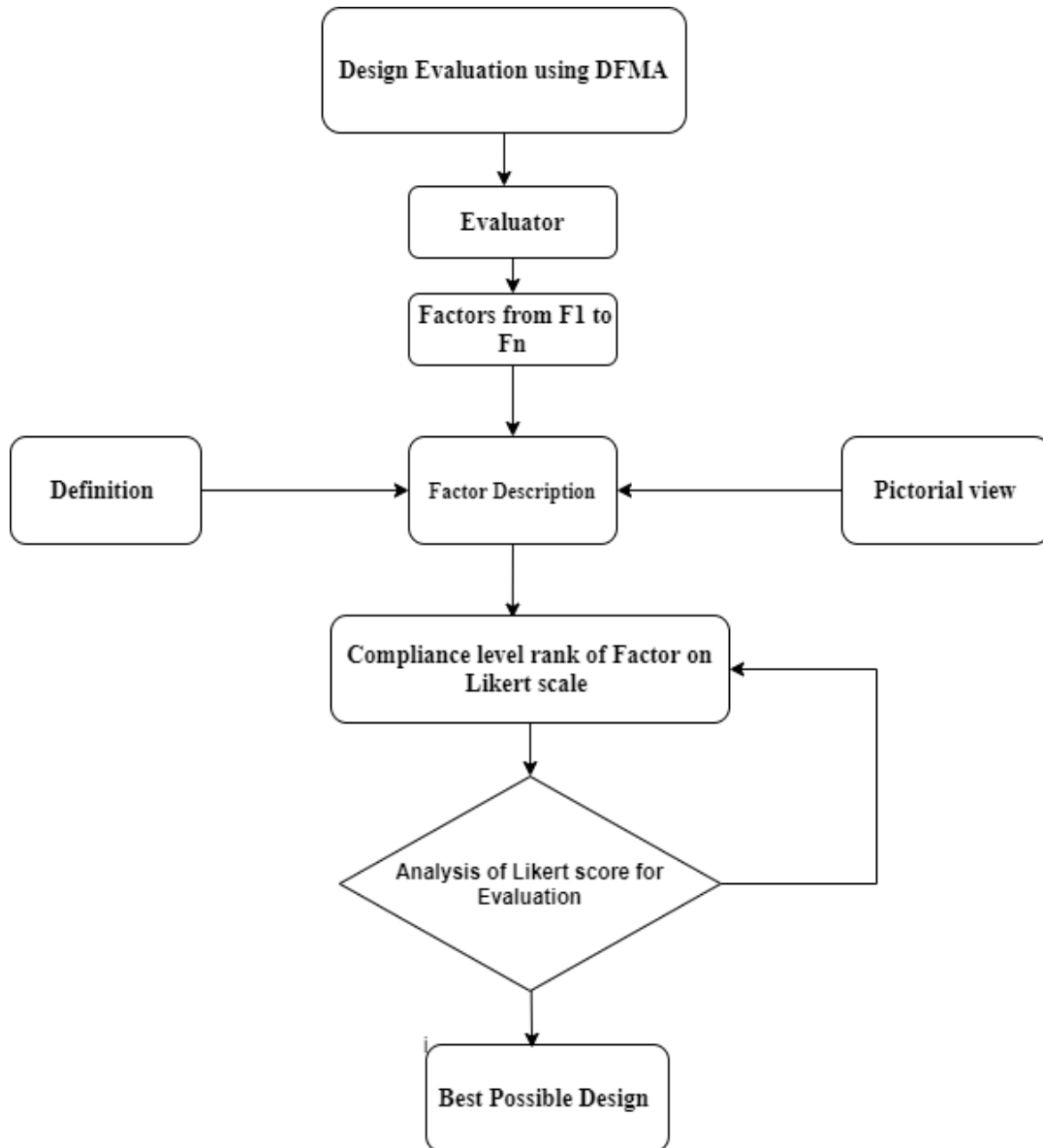


Figure 6 Framework of BIM-DE4PB Plugin Development

The development environment that is utilized for model development comprises of numerous BIM-based software's that includes Autodesk Revit Architecture 2022 (BIM Software), Visual Studio.Net (Software Development Environment); and Structured Query Language (SQL) Server (Database Management System). Autodesk Revit Architecture 2022 is utilized due to top notch

visual introductions and its simplicity of availability with outer information base (Riaz et al. 2014). The justification choosing Autodesk Revit as a BIM stage for module improvement is that it is generally utilized and gives a rich API stage (Davila Delgado et al., 2017). Revit API gives an exceptionally adaptable .NET Software Development Toolkit for UI and programming rationale advancement. A usually utilized multi-worldview programming language (i.e., JavaScript) was utilized to make the model utilizing Microsoft visual studio. An information association was set up between Revit Architecture programming and SQL Server utilizing the API of Revit programming.

3.7.1 The DE4PB Ribbon.

The plugin is created by using .Net Framework 4.7 and JavaScript programming language using visual studio community. It is comprised of the following panel and buttons as shown in Figure 7 & Figure 8.

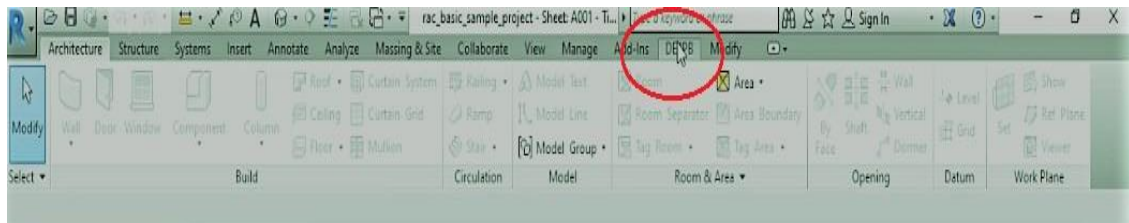


Figure 7 Design evaluation plugin BIM-DE4PB Rebin

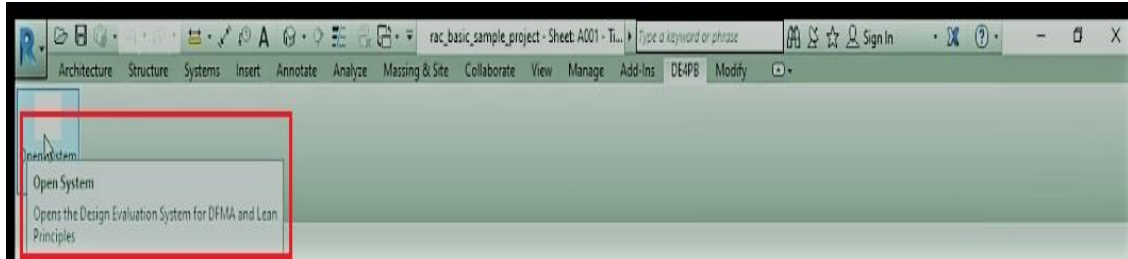


Figure 8 BIM-DE4PB Ribbon Panel and Evaluation button

3.7.2 Design Evaluator Details.

The evaluator will input his/her details so that data can be counter checked and the knowledge of evaluator about the said field verified. System will generate a unique random ID for each evaluator. The ID (identity number) will be generated each time user will start evaluation as shown in the Figure 9.

Figure 9 BIM-DE4PB Evaluator Registration Detail

Only after putting evaluator detail, he can proceed for next step of evaluation.

3.7.3 Selection of Design Evaluation Method among DFMA & Lean Principles.

Once design evaluator will proceed further, he has to choose evaluation method between DFMA (Design for Manufacture and assembly) and Lean principles. upon moving cursor on one the method evaluator will see the brief description of the methods as shown in Figure 10.

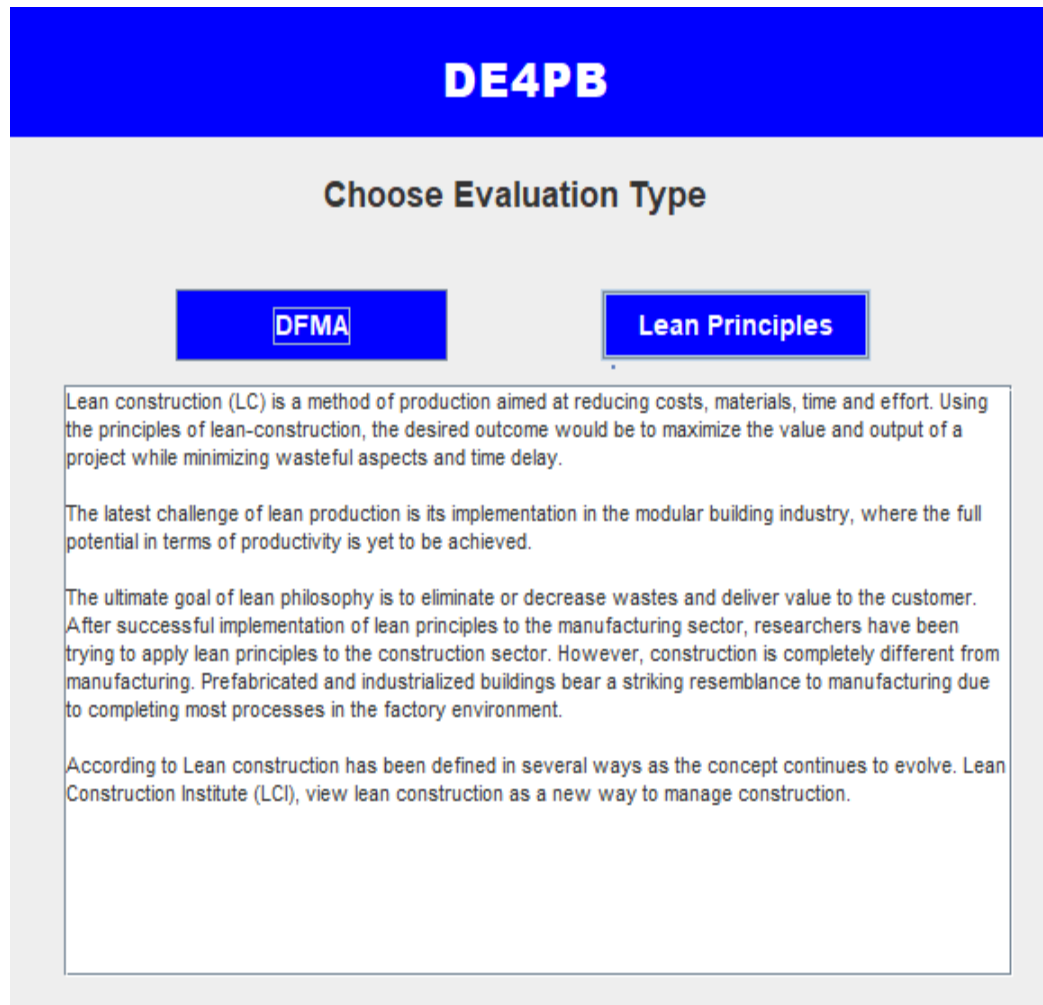


Figure 10 BIM-DE4PB Evaluation Method Description

3.7.4 Evaluation of design by Ranking of compliance level of factors.

After reading the description of the method, evaluator will select the evaluation method. The evaluator checks the compliance level of each factor as shown in the Figure 11 for the design that he has made in Revit. None of the factor should be less than three (Neutral) on Likert scale. Evaluator will read the description of each factor which consists of brief explanation and pictorial view. This method is called heuristic evaluation.

BIM Based Design Evaluation using Lean Principles

Dependencies between design activities.

A variety of activity dependencies exists, and activity relationships are categorized in different ways. The four main types of activity dependencies include Finish-to-Start (FS), Start-to-Start (SS), Start-to-Finish (SF), and Finish-to-Finish (FF). Every project has dependencies, which Max Wideman's Glossary defines as the "relationships between products or tasks", i.e. tasks that require input from other tasks to be completed, or activities that can't start until a previous activity is done.

Mubarak, S. A. (2010) "Construction project scheduling and control", (2010)

F1L	F2L	F3L	F4L	F5L
F6L	F7L	F8L	F9L	F10L
F11L	F12L	F13L	F14L	F15L
F16L	F17L	F18L	F19L	F20L
F21L	F22L	F23L	F24L	F25L
F26L	F27L	F28L	F29L	

☐ Very Unsatisfied
☐ Unsatisfied
☐ Neutral
☐ Satisfied
☐ Very Satisfied

Back
Submit
Next

Figure 11 BIM Based Design Evaluation using Lean Principles

3.7.5 Heuristic evaluation method

Heuristic evaluation is the most usually utilized ease of use review strategy. It became well known in the mid-1990s as a result of its speed, efficiency, and simplicity of execution. It very well may be accomplished with just 4–5 evaluators utilizing a restricted arrangement of standards to identify a high extent of ease-of-use issues (Ekanayake & Ofori, 2004). In heuristic assessment, every evaluator investigates the planned framework or plugin alone and passes judgment on its consistency as per a bunch of ease-of-use standards (Baghdadi et al., 2020). Heuristic assessments can be executed rapidly and helpfully through a skilled pool of evaluators and the most notable heuristic standards are the 10 created by Nielsen (1992). Assessing the current practice in heuristic assessments showed that Nielsen's heuristic assessment strategy has been concentrated according to alternate points of view during the last decade, like the number of evaluators (*Applying Heuristic Evaluation in Usability Testing*, 2018.), evaluator impact) and issue seriousness (Ciardiello et al., 2020).

In addition, there are additionally heuristic assessments joined with other ease of use assessment strategies for a more productive and viable convenience examination, depiction, and detailing measure, for example, heuristic assessments joined with observational testing at various periods of the advancement cycle (Zhang et al., 2020) as shown in Figure 12. heuristic assessments with a progressive construction instead of Nielsen's level design of 10 heuristics (Andre et al., 2001) and heuristic assessments joined with the Systematic Usability Evaluation (SUE) technique to beat the issues experienced in Nielsen's heuristics through a more engaged ease of use review strategy. Besides, other heuristic assessment techniques have been likewise proposed in various examinations, like three classifications of heuristic application, lead, follow, and move (LF&G), which helped the formation of checklist and assessment structures for use across all phases of development of the framework (Kamper, 2002), and broadened convenience heuristics by presenting 3D visual environment (VE) particular standards (Neelamkavil, 2009)



Figure 12 Heuristic Evaluation Benefits

3.7.6 Design Evaluation Report in PDF format.

In last phase evaluator can print his report in PDF format and will see the results. Those factors whose score is less than three (Neutral) on Likert scale will be marked Red. BIM based report has Note at bottom containing the message that red marked factors need to be improve in the design.

Evaluation of the System

4.1 EVALUATION RESULTS

Industry experts were given the execution of BIM-DE4PB on a genuine model for development of the design of pre-assembled constructing components. Results of the expert's reviews sessions are discussed in the following subsections.

4.1.1 Result of Questionnaire

Figure 13 show the expert's reaction for questions identified with the need of BIM-DE4PB for design improvement and assessment in construction industry, ease of use of BIM-DE4PB interface, possible application of BIM-DE4PB, and generally effectiveness to further development of design of prefabricated building element at initial stage of project.

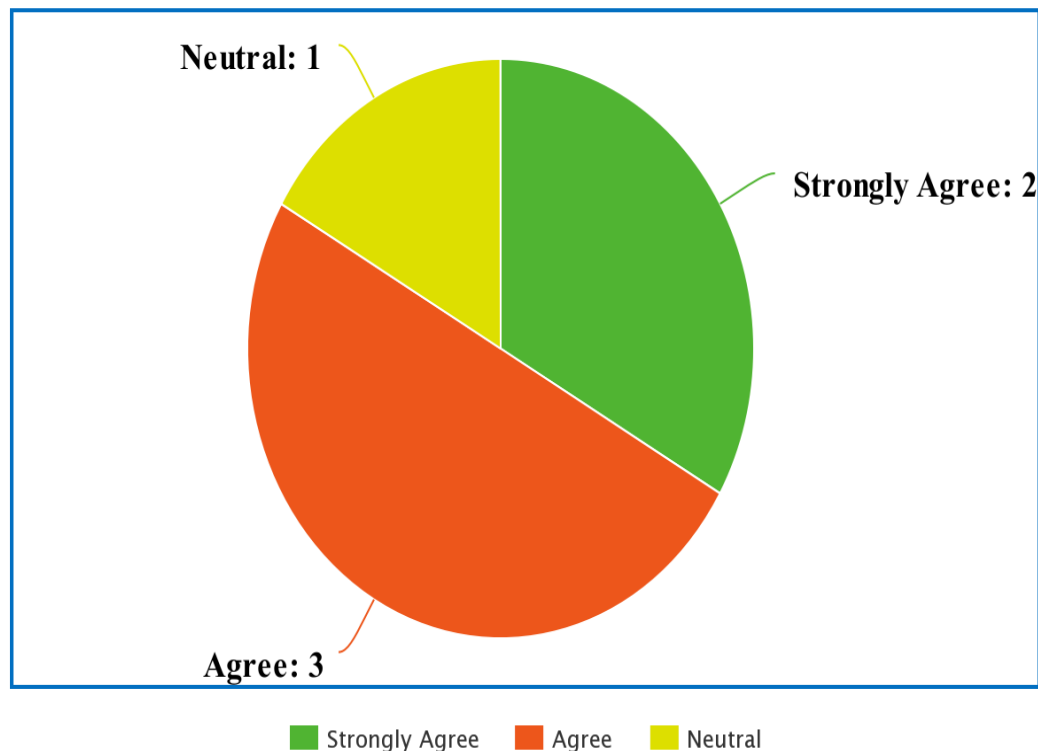


Figure 13 Need of BIM-DE4PB

Out of 6 experts of prefabricated construction technology 83 % (50% strongly agree and 33 % agree) thought that there is need of such system for evaluation of prefabricated building element design. While 16% experts have

remained un-decided as shown in Figure 14. Encouragingly, none of the experts disagreed with its need in construction industry.

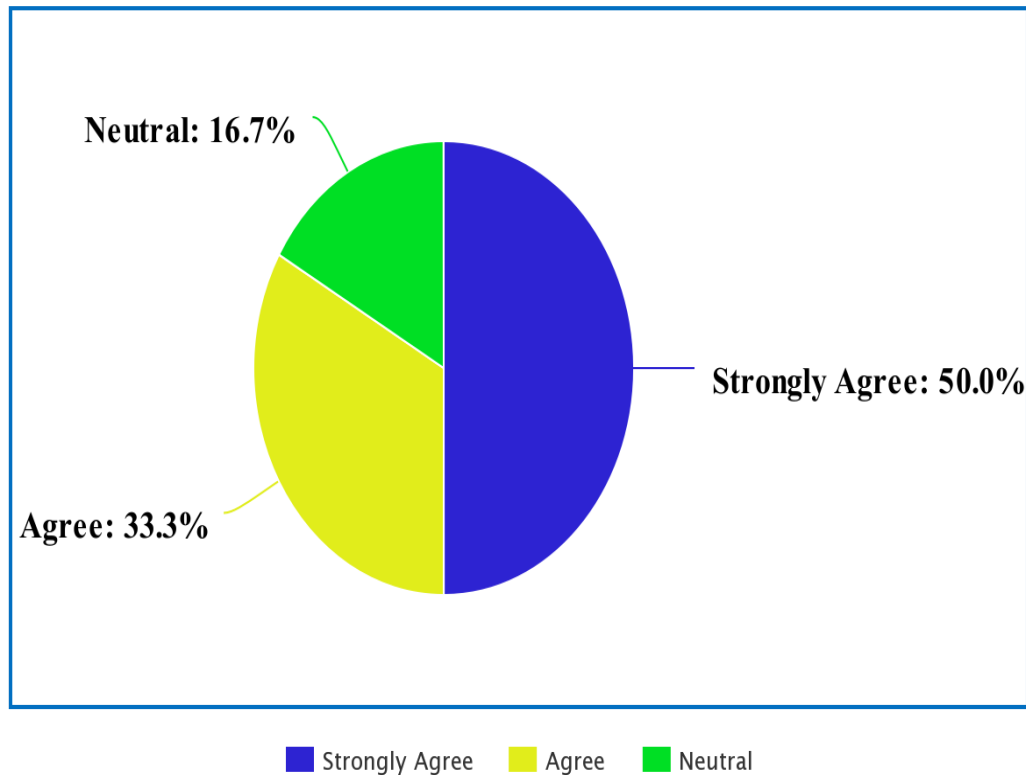


Figure 14 Experts Response on Usability of BIM-DE4PB System

Out of 6 experts of prefabricated construction technology 83 % (50% strongly agree and 33 % agree) thought that overall system can use very easily. While 16% experts have remained un-decided as shown in Figure 15. This high ratio shows that particular attention should be given in educating people with working of BIM.

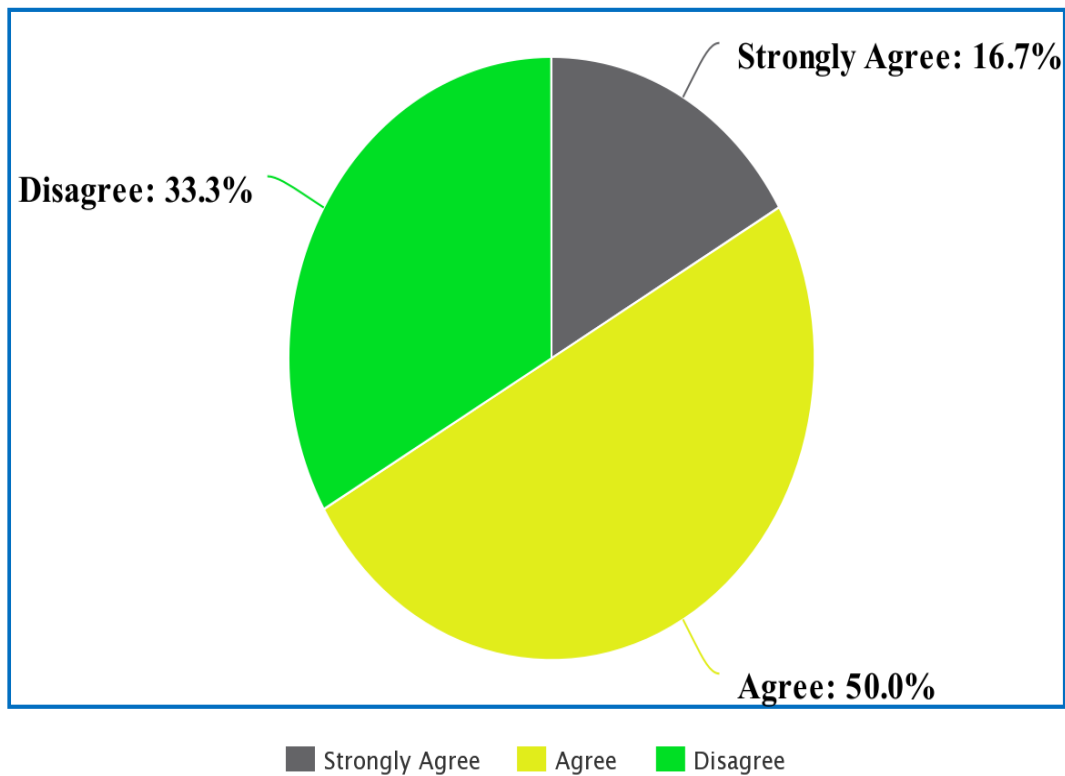


Figure 15 Experts Response on Implementation of BIM-DE4PB system

Overall, 66% experts were of opinion that the system can be implemented for projects, whereas because of the existing traditional practices in construction industry and absence of bylaws 33% thought its implementation is not possible. Likewise, some experts commented that implementation is possible by taking certain measures, as many clients and Government bodies have made compulsory to submitting the BIM model of design at the initial stage of the project.

4.2 EXPERTS FEEDBACK ON POTENTIAL OF BIM-DE4PB IN RESOLVING IDENTIFIED DESIGN ISSUES AT INITIAL STAGE

Table 6 shows the experts' opinion about the factors listed below if followed can improve the design of prefabricated building elements by the proposed system? These issues can be ranked based on their importance on a 5-point Likert scale.

(where 1=very unsatisfied, 2=unsatisfied, 3= Neutral, 4=Satisfied level, 5=extremely satisfied)

Table 6 Experts scores for importance of each factor for BIM-DE4PB

S. No	Design Evaluation factors from Lean Principles	Average Score	RII Value	Overall Ranking
1	Dependencies between design activities.	4.5	0.9	1
2	Retaining all details of building components (strength, material, cost, dimensions, etc)	4.5	0.9	1
3	Time required for assembling of the building element	3.83333	0.76667	5
4	Access and vision restrictions for Manual assembly	3.83333	0.76667	5
5	Modularize ability of the building element	3.5	0.7	7
6	Size of the building element	3.5	0.7	7
7	Number of different components in building elements	3.66667	0.73333	6
8	ship-ability	3.5	0.7	7
9	Ease of fabrication	4.16667	0.83333	3
10	Symmetry of design	3.5	0.7	7
11	Less waste generation	3.5	0.7	7
12	Designer Knowledge about possibilities of manufacturing, outsource options	3.5	0.7	7
13	Function and Performance requirement of the particular Building element	3.66667	0.73333	6
14	Amount of storage required at the construction site.	4	0.8	4
15	The requirement for High-Speed Automatic Assembly and Robot Assembly	3.83333	0.76667	5
16	Reuse Ability of building component	3.5	0.7	7
17	Easy handling of Building Element	3.33333	0.66667	8
18	The need of at once addition of different parts that should be kept in alignment	3.83333	0.76667	5
19	Coordination mechanism	4.33333	0.86667	2
20	Tangling and Nesting provisions for different building elements for better handling during assembly	3.83333	0.76667	5

21	Applied loads and resistance of the material.	4.16667	0.83333	3
22	The layered (top-down assembly) design approach	3.66667	0.73333	6
23	Specialized workforce requirement	3.66667	0.73333	6
24	Effect of assembly layout on Acquisition Times	4.16667	0.83333	3
25	Availability of building element as per need	4.16667	0.83333	3
26	Market demand	3.66667	0.73333	6
27	Use of moulded-in identification instead of attached labels.	4	0.8	4
28	Expected Level of Production	3.83333	0.76667	5
29	Salvage value of Building component	3	0.6	9

Table 7 shows the opinion taken about factor of DFMA (design for manufacture and assembly).

Table 7 Design Evaluation Factors from DFMA

S. No	Design Evaluation Factors from DFMA	Average Score	RII Value	Overall Ranking
1	Complexity of geometry	4.5	0.9	1
2	Environmental Friendliness	4.5	0.9	1
3	Health and safety risk involved in design option	4.16667	0.83333	3
4	Services (plumbing, electrical, HVAC) separation from structure or serviceability requirement of design	4.33333	0.86667	2
5	Importance of conceptual stage design	3.83333	0.76667	5
6	Design performance for exposure to Moisture (Fresh Water, Sea Water) and drainage	3.5	0.7	7
7	Higher initial capital cost / Manufacturing cost	3.66667	0.73333	6
8	Simplified design /Use the simplest cross-sectional shape	3.83333	0.76667	5
9	Selection of material and product design relation	3.66667	0.73333	6
10	Manufacturing ability of the existing assembly line.	3.16667	0.63333	9

11	Heat conductivity of material required for particular building elements	4.16667	0.83333	3
12	Expertise Required to design particular element	3.83333	0.76667	5
13	Tolerance tightness level and surface roughness requirement	4	0.8	4
14	Ease of disassembly the building element	4	0.8	4
15	The need for holding parts down to keep up with their direction during control of the sub-assembly	4.16667	0.83333	3
16	Basic knowledge of engineering requirement of particular design i.e., material, manufacturing physics	3	0.6	10
17	Does Information regarding machine capabilities and processing need at the initial design stage matter?	3.66667	0.73333	6
18	Use sub-assemblies	3.66667	0.73333	6
19	Cost data for different materials for different building components.	3.5	0.7	7
20	The durability of material required for particular building elements design	3.5	0.7	7
21	Visibility and accessibility of parts prone to wear or failure	3.33333	0.66667	8
22	Recycle Ability of the Material	4	0.8	4
23	Client interest or possible financial benefit of design	3.33333	0.66667	
24	Function of building element, operational action, and maintenance requirement	2.83333	0.56667	11
25	Customer requirement importance	3.5	0.7	7
26	Reliability of individual components to make the design more reliable.	3.83333	0.76667	5
27	Heat capacity of material required for particular building elements	4.16667	0.83333	3
28	Bye law's requirement of design from development authorities	4	0.8	4
29	The ability to use general-purpose tooling and process	3.5	0.7	7
30	Design compliance with standards	3.66667	0.73333	6
31	Dimensions requirement	3.66667	0.73333	6
32	Standardization ability of Building element design	3.66667	0.73333	6

33	Design the building element in such a way that as many manufacturing operations as possible can be performed without repositioning it.	3.5	0.7	7
34	Upgrade ability provisions of design	2.83333	0.56667	
35	Number of parts in Welded assemblies	3.16667	0.63333	9
36	Methods and technique required to manufacture the product	3.66667	0.73333	6
37	In-Service Temperature requirement of particular building element	3.66667	0.73333	6
38	Designing parts such that they cannot be inserted incorrectly or zero fitting problem	3.16667	0.63333	9
39	The type of surface texture customer needs	3.83333	0.76667	5
40	which standard shape is available and easy to make for particular building element	3.33333	0.66667	8
41	Appearance of element or how attractive the design of product.	3.33333	0.66667	8
42	Jamming of parts during installation	3.33333	0.66667	8
43	Scaffolding needs for the particular design of elements	3.5	0.7	7
44	Self-tapping screws use for particular building element	3	0.6	10
45	Design of the product having part is located before it is released.	3.33333	0.66667	8
46	Manual Handling requirement of design.	3.33333	0.66667	8
47	Transparency and Colour Requirements for kitchen and bathroom	3.16667	0.63333	9
48	User-friendliness or ergonomics of the design	3.16667	0.63333	9
49	Use of snap fits	3.66667	0.73333	6
50	Use of the number of fasteners or connectors	3.5	0.7	7
51	Scrap rates of design	3.5	0.7	7
52	Appropriate space between connecting holes.	3	0.6	10
53	Weight consistent with strength and stiffness requirements	3.33333	0.66667	8
54	Edge preparation required for a particular building element.	3.33333	0.66667	8

55	Use a structural grid base design	3.16667	0.63333	9
56	Use mechanical connections rather than chemical ones	2.83333	0.56667	11
57	Score of Manufacturing ability during design phase	2.83333	0.56667	11
58	The importance of Historic design information	2.83333	0.56667	11
59	Amount of resistance to insertion and requirement of chamfer	3.16667	0.63333	9
60	Designing of building component in which components need two hands for Manipulation.	3.16667	0.63333	9
61	Setting up an information base that comprises assembly times and cost factors for different design circumstances and production conditions.	3.16667	0.63333	9
62	Metal stamping process requirements for particular elements	2.66667	0.53333	12
63	Suitable material for stamping	2.33333	0.46667	13

4.3 DISCUSSION ON EXPERTS SCORE

During semi-structured interviews, expert's panel expressed their views on the stated problematic issues. Key comments are discussed below in groups.

In Need of the BIM-DE4PB group, the average score of all industry experts indicated that most of the experts scored (Agree and strongly agreed), consequently BIM-DE4PB is able to evaluate the design issues of prefabricated building elements at the conceiving stage of project.

In *useability of the BIM-DE4PB* group, the mean score of all experts indicated that most of the experts scored (Agree and strongly agreed). Thus BIM-DE4PB plugin is very easy to use. The graphical user interface of the plugin has been designed in such a way that clients can undoubtedly see each progression to assess the design of prefabricated building element.

As for *Implementation group of the BIM-DE4PB*, the average score of all experts indicated that most of the experts scored (Agree and strongly agreed), hence BIM-DE4PB is able to implement in the construction industry. While 33% has

highlighted the issue that in local construction industry of Pakistan BIM already trained personnel are not present. Professionals want to follow the traditional design modification technique's but on the other hand BIM-DE4PB will provide the user an integrated visual based design evaluation platform to make the design more efficient.

In *DFMA (design for manufacture and assembly) group*, the average score of all participants indicated that all factors are playing important role towards evaluation of prefabricated design through BIM-DE4PB.

In general, BIM-DE4PB can be considered as a successful and constructive tool to solve the problem of BIM based design evaluation using DFMA & lean principles to a great extent.

4.4 FEEDBACK ON BARRIERS IN BIM-DE4PBE IMPLEMENTATION

Table 8 sums up key barriers to the powerful execution of BIM-DE4PB in the construction industry. These hurdles were distinguished by the prefabricated industry experts during semi-organized meetings for the evaluation of the BIM - DE4PB tool.

Table 8 Implementation Barriers for BIM -DE4PB

Barriers	Key Comments
Absence of BIM knowledge	A great deal of endeavours should be gone through on instructing individuals with the information on BIM.
Cost	Proper training and costly equipment are required for using BIM and BIM-DE4PB Outsourcing to 3rd party as BIM specialist is a costly solution
Reluctance for innovative ideas in traditional construction practice	It is truly challenging to carry out the new BIM-based system where everybody isn't prepared. adoption is troublesome because of related risks with new innovative technology in construction industry.

Non-serious attitude towards managing with Design evaluation at initial stage	Design evaluation at conceiving stage of project is very essential. But lot of industry professional want to save the cost at initial design phase which they have to pay after completion.
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CONCLUSIONS AND RECOMMENDATIONS

5.1 INTRUCTION

This chapter summarize the research by stating the conclusions, recommendation, and conclusion.

5.2 CONCLUSION

Modern BIM based technologies and gears offer valuable significance in improving the construction engineering and management practices.this study integrate the DFMA and lean principles for design evaluation of any prefabricated building element.

The Information and Communication Technology (ICT) use has immensely worked with the industry experts in making more noteworthy undertaking progress and high usefulness for minimal price. Building Information Modeling (BIM), which has caused a change in outlook in the construction industry for improving the presentation of practically all parts of construction engineering & management like sustainable construction, construction facility management, construction safety management and risk management etc.

A systematic methodology has adopted to identify the design evaluation factors through extensive literature review for prefabricated building elements using DFMA and lean principles. BIM itself do not have the provision of design evaluation of prefabricated building elements but due to open API provided by the BIM based software, various type of BIM based prototype can be developed. Hence a prototype (Plugin in Revit) has developed for Design evaluation of prefabricated building elements using DFMA and lean principles named BIM based Design Evaluation for Prefabricated Buildings (BIM-DE4PB). The system can apply to evaluate the design of any Prefabricated Building Element.

A commonly use programming language (JavaScript) was utilized to develop prototype using Microsoft visual studio. BIM-DE4PB was used to generate the BIM based Design Evaluation report, which can be saved in PDF format for further use.

The BIM-DE4PB is able to evaluate the design of any prefabricated building element at the conceiving stage of the project. The BIM based system help the designer to design most efficient design.

Some of the barriers in BIM-DE4PB implementation, highlighted by the experts, are lack of BIM knowledge, costly solution, resistance to innovation in traditional construction practices, non-serious attitude towards dealing with design issues. The specialists likewise proposed measures to work on the convenience and adequacy of the developed BIM plugin (BIM-DE4PB).

5.3 RECOMMENDATIONS

As the created BIM-DE4PB design evaluation plugin is validated through experts, it should be implemented in true spirit on prefabricated construction projects from initiation phase of the project to evaluate the design of prefabricated building elements. A lot of design issues regarding designing, assembling, industrial waste creation, suitability and many more forthcoming issues can be easily avoided. BIM enable plugin will provide the real time visual of design issue and their instant solution. Different evaluators input can be integrated, and design can be improved. The developed system will provide more lucidity and transparency, moreover it provides satisfaction of all stakeholders at initial stage which will further reduce design communication issues.

5.4 LIMITATIONS

One of the limitations of the system is that only expert of DFMA and lean Principle can use the BIM-DE4PB more efficiently.

5.5 FUTURE RESEARCH DIRECTIONS

Improvement can be made in the BIM-DE4PB system further extent its application. A study can be made to further find financial benefit of this BIM based design evaluation system. Sustainability guidelines of Building design can also be integrated with the system can be explored

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APPENDIX-A

Evaluation of BIM-DE4PB from Industry Experts



The purpose of this exercise is to evaluate the system developed by the Construction Engineering & Management department of National University of Sciences and Technology (NUST), H-12 sector Islamabad, Pakistan, for the construction industry to BIM based design Evaluation of Prefabricated Building elements using DFMA and Lean Principles.

All discussions will be kept anonymous, and the information provided will solely be used only for academic purposes. The discussion will be voice recorded, but you can choose to answer the questions in written form depending on your ease and preferable mode.

Thank you for your Cooperation.

Mubashir Islam

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Pakistan.

The experts review will be comprised of two sections.

Section 1: Profile of Expert.

Section 2: Evaluation questions and discussions.

SECTION: 1 (Profile of Expert)

Please mention the following:

Organization Type	
Designation in the Company or Organization	
Experience (in Years)	

Please choose your answer on the given scale.

(1=strongly disagree, 2= disagree, 3= neutral, 4= agree, 5= strongly agree)

1. There is a need to introduce such a system in the Construction industry.

1	2	3	4	5
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The proposed system would be easy to use.

1	2	3	4	5
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2. The proposed system is implementable. (Please specify the possible barriers you feel in its implementation)

1	2	3	4	5
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Please describe your comments for the barriers in its implementation

3. To what extent, do you think the Factors listed below if followed can improve the design of prefabricated building elements by the proposed system? The improvement in design can be ranked based on their importance on a 5-point Likert scale.

(where 1=very unsatisfied, 2=unsatisfied, 3= Neutral, 4=Satisfied level , 5=extremely satisfied)

S. No	Factors from DFMA for Design optimization and Evaluation					
1	Complexity of geometry	1	2	3	4	5
2	Environmental Friendliness	1	2	3	4	5
3	Health and safety risk involved in design option	1	2	3	4	5
4	Services (plumbing, electrical, HVAC) separation from structure or serviceability requirement of design	1	2	3	4	5
5	Importance of conceptual stage design	1	2	3	4	5

6	Design performance for exposure to Moisture (Fresh Water, Sea Water) and drainage	1	2	3	4	5
7	Higher initial capital cost / Manufacturing cost	1	2	3	4	5
8	Simplified design /Use the simplest cross-sectional shape	1	2	3	4	5
9	Selection of material and product design relation	1	2	3	4	5
10	Manufacturing ability of the existing assembly line.	1	2	3	4	5
11	Heat conductivity of material required for particular building elements	1	2	3	4	5
12	Expertise Required to design particular element	1	2	3	4	5
13	Tolerance tightness level and surface roughness requirement	1	2	3	4	5
14	Ease of disassembly the building element	1	2	3	4	5
15	The necessity for holding parts down to maintain their orientation during manipulation of the sub-assembly.	1	2	3	4	5
16	Basic knowledge of engineering requirement of particular design i.e., material, manufacturing physics	1	2	3	4	5
17	Does Information regarding machine capabilities and processing need at the initial design stage matter?	1	2	3	4	5
18	Use sub-assemblies	1	2	3	4	5
19	Cost data for different materials for different building components.	1	2	3	4	5
20	The durability of material required for particular building elements design	1	2	3	4	5
21	Visibility and accessibility of parts prone to wear or failure	1	2	3	4	5
22	Recycle Ability of the Material	1	2	3	4	5
23	Client interest or possible financial benefit of design	1	2	3	4	5
24	Function of building element, operational action, and maintenance requirement	1	2	3	4	5
25	Customer requirement importance	1	2	3	4	5
26	Reliability of individual components to make the design more reliable.	1	2	3	4	5
27	Heat capacity of material required for particular building elements	1	2	3	4	5
28	Bye law's requirement of design from development authorities	1	2	3	4	5
29	The ability to use general-purpose tooling and process	1	2	3	4	5
30	Design compliance with standards	1	2	3	4	5
31	Dimensions requirement	1	2	3	4	5
32	Standardization ability of Building element design	1	2	3	4	5

33	Design apart so that as many manufacturing operations as possible can be performed without repositioning it.	1	2	3	4	5
34	Upgrade ability provisions of design	1	2	3	4	5
35	Number of parts in Welded assemblies	1	2	3	4	5
36	Methods and technique required to manufacture the product	1	2	3	4	5
37	In-Service Temperature requirement of particular building element	1	2	3	4	5
38	Designing parts such that they cannot be inserted incorrectly or zero fitting problem	1	2	3	4	5
39	The type of surface texture customer needs	1	2	3	4	5
40	which standard shape is available and easy to make for particular building element	1	2	3	4	5
41	Appearance of element or how attractive the design of product.	1	2	3	4	5
42	Jamming of parts during installation	1	2	3	4	5
43	Scaffolding needs for the particular design of elements	1	2	3	4	5
44	Self-tapping screws use for particular building element	1	2	3	4	5
45	Design of the product having part is located before it is released.	1	2	3	4	5
46	Manual Handling requirement of design.	1	2	3	4	5
47	Transparency and Colour Requirements for kitchen and bathroom	1	2	3	4	5
48	User-friendliness or ergonomics of the design	1	2	3	4	5
49	Use of snap fits	1	2	3	4	5
50	Use of the number of fasteners or connectors	1	2	3	4	5
51	Scrap rates of design	1	2	3	4	5
52	Appropriate space between connecting holes.	1	2	3	4	5
53	Weight consistent with strength and stiffness requirements	1	2	3	4	5
54	Edge preparation required for a particular building element.	1	2	3	4	5
55	Use a structural grid base design	1	2	3	4	5
56	Use mechanical connections rather than chemical ones	1	2	3	4	5
57	Score of Manufacturing ability during design phase	1	2	3	4	5
58	The importance of Historic design information	1	2	3	4	5
59	Amount of resistance to insertion and requirement of chamfer	1	2	3	4	5
60	Designing of building element in which Parts Requiring two hands for Manipulation	1	2	3	4	5

61	Establishing a database that consists of assembly times and cost factors for various design situations and production conditions.	1	2	3	4	5
62	Metal stamping process requirements for particular elements	1	2	3	4	5
63	Suitable material for stamping	1	2	3	4	5

S. No	Design optimization and Evaluation factors from Lean Principles					
1	Dependencies between design activities.	1	2	3	4	5
2	Retaining all details of building components (strength, material, cost, dimensions, etc)	1	2	3	4	5
3	Time required for assembling of the building element	1	2	3	4	5
4	Access and vision restrictions for Manual assembly	1	2	3	4	5
5	Modularize ability of the building element	1	2	3	4	5
6	Size of the building element	1	2	3	4	5
7	Number of different components in building elements	1	2	3	4	5
8	ship-ability	1	2	3	4	5
9	Ease of fabrication	1	2	3	4	5
10	Symmetry of design	1	2	3	4	5
11	Less waste generation	1	2	3	4	5
12	Designer Knowledge about possibilities of manufacturing, outsource options	1	2	3	4	5
13	Function and Performance requirement of the particular Building element	1	2	3	4	5
14	Amount of storage required at the construction site.	1	2	3	4	5
15	The need for High-Speed Automatic Assembly and Robot Assembly	1	2	3	4	5
16	Reuse Ability of building component	1	2	3	4	5
17	Easy handling of Building Element	1	2	3	4	5
18	The requirement of simultaneous addition of several parts that must be kept in alignment	1	2	3	4	5
19	Coordination mechanism	1	2	3	4	5
20	Tangling and Nesting provisions for different building elements for better handling during assembly	1	2	3	4	5
21	Applied loads and resistance of the material.	1	2	3	4	5
22	The layered (top-down assembly) design approach	1	2	3	4	5
23	Specialized workforce requirement	1	2	3	4	5

24	Effect of assembly layout on Acquisition Times	1	2	3	4	5
25	Availability of building element as per need	1	2	3	4	5
26	Market demand	1	2	3	4	5
27	Use of moulded-in identification instead of attached labels.	1	2	3	4	5
28	Expected Level of Production	1	2	3	4	5
29	Salvage value of Building component	1	2	3	4	5

4. Overall, the proposed BIM based system would be effective for improvement of design of prefabricated building elements at conceiving stage of project. (Scale: 1=strongly disagree, 2= disagree, 3= neutral, 4= agree, 5= strongly agree)

1	2	3	4	5
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